

- 2 (a) (i) no forces (of attraction or repulsion) between atoms / molecules / particles B1 [1]
- (ii) sum of kinetic and potential energy of atoms / molecules M1
due to random motion A1 [2]
- (iii) (random) kinetic energy increases with temperature M1
no potential energy
(so increase in temperature increases internal energy) A1 [2]
- (b) (i) zero A1 [1]
- (ii) work done = $p\Delta V$ C1
 $= 4.0 \times 10^5 \times 6 \times 10^{-4}$
 $= 240 \text{ J}$ (ignore any sign) A1 [2]

(iii)

change	work done / J	heating / J	increase in internal energy / J
P → Q	+240	−600	−360
Q → R	0	+720	+720
R → P	−840	+480	−360

(correct signs essential)

(each horizontal line correct, 1 mark – max 3)

B3 [3]

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